Appendix to "Online Dispatch of Multi-Duration Storage via Spectral Decomposition of Locational Marginal Prices"

I. Proofs

Unprefixed equation numbers refer to the letter. *A. Proof of Theorem 1.* We price the network at the LMPs, so that each unit solves its own problem, frequency by frequency. First, dualizing (1b) and (1g) in (3) prices the injection at bus n at the LMP λ_n. The dualized Lagrangian is separable, so at a finite-cost saddle point, unit i solves, with its bus LMP $\lambda = \lambda_{n(i)}$ fixed,

$$\min_{p_i} V_i = \int \left(c_i^\tau p_i^2 + \kappa_i \varepsilon_i^2 - \lambda p_i\right) \mathrm{d}t \;\text{ s.t. }\; \dot{\varepsilon}_i = -p_i. \tag{A1}$$

Then, by Parseval's theorem and $\hat{\varepsilon}_i = -\hat{p}_i/\mathrm{j}\omega$,

$$V_i = \tfrac{1}{2\pi}\int \left[a_i|\hat{p}_i|^2 - \Re(\hat{\lambda}^* \hat{p}_i)\right] \mathrm{d}\omega, \tag{A2}$$

whose integrand is a separate quadratic in $\hat{p}_i(\omega)$ for each ω. Setting its derivative to zero gives the output at each frequency

$$\hat{p}_i = \hat{\lambda}/(2a_i), \quad \omega \neq 0. \tag{A3}$$

This solution has finite energy, as $|\hat{p}_i| \le |\hat{\lambda}|/(2c_i^\tau)$ and $|\hat{\varepsilon}_i| \le |\hat{\lambda}|/(4\sqrt{c_i^\tau \kappa_i})$, and a constant in λ does not change it, as $a_i(0) = \infty$. Finally, for the multiplier μ_i of $\dot{\varepsilon}_i = -p_i$, stationarity gives $\mu_i = \lambda - 2c_i^\tau p_i$. Substituting (A3) with $c_i^\tau/a_i = \tau_i^2\omega^2/(1+\tau_i^2\omega^2)$ gives the low-pass filter of the bus LMP with corner $1/\tau_i$

$$\hat{\mu}_i = \hat{\lambda} - 2c_i^\tau \hat{p}_i = \hat{\lambda}/(1+\tau_i^2\omega^2). \tag{A4}$$

As $1/(1+\tau_i^2\omega^2)$ is the transform of $e^{-|t|/\tau_i}/(2\tau_i)$, (A4) is the convolution (4) in time, which completes the proof. ∎

B. Proof of Corollary 1. We insert (A3) into the power balance and solve for the single price, and then show that no causal policy attains (5). The generator sets its marginal cost to the price, $\lambda = 2qs$, so $\hat{s} = \hat{\lambda}/(2q)$. Substituting this and (A3) into the balance $\sum_i \hat{p}_i + \hat{s} = \hat{x}$ gives the single price

$$\hat{\lambda} = 2\hat{x}\big/\big(q^{-1} + \textstyle\sum_{i'} a_{i'}^{-1}\big), \tag{A5}$$

and inserting (A5) back into $\hat{p}_i = \hat{\lambda}/(2a_i)$ and $\hat{s} = \hat{\lambda}/(2q)$ gives (5). This is the optimum of the strictly convex problem (3), as $0 \le H_i \le 1$ and $H_i = q\omega^2/\kappa_i + O(\omega^4)$ near $\omega = 0$, so p_i and ε_i have finite energy. For the split, $H_i \propto 1/a_i$, so for $c_i^\tau < c_j^\tau$ and $\kappa_i > \kappa_j$, $H_i > H_j$ if and only if $\omega > \omega_{ij} = \sqrt{(\kappa_i - \kappa_j)/(c_j^\tau - c_i^\tau)}$. Ordered crossovers $\omega_{12} > \cdots > \omega_{N-1,N}$ thus give unit k the largest share on $(\omega_{k,k+1}, \omega_{k-1,k})$. *Non-causality.* A causal filter with a real frequency response is a constant gain, which H_i is not. Let h be the impulse response of a BIBO-stable causal LTI system with a real frequency response. Stability makes h a finite measure, a real response makes h even, and an even h supported on $[0,\infty)$ is a point mass. Then, the kernel h_i of H_i is even, as H_i is real, and is not a point mass, as $H_i(0) = 0$ and $H_i \not\equiv 0$. It thus has mass on $(-\infty, 0)$, i.e., H_i responds to future net load, which no causal policy does, so no causal filter attains (5). ∎

C. Proof of Theorem 2. We derive (6), write the proposed dispatch as the hindsight optimum plus a filtered prediction error, and project onto causal policies. Throughout, the leak $\dot{\varepsilon}_i = -\nu\varepsilon_i - p_i$, $\nu > 0$, replaces κ_i/ω^2 in a_i by $\kappa_i/(\omega^2+\nu^2)$; H, M, and μ are those of this model, M is bounded, and (7) is finite for integrable Φ_δ. Without it, (7) needs, e.g., $\Phi_\delta = O(\omega^2)$ at $\omega = 0$, as a persistent error drifts the SoC. *Cost decomposition (6).* We complete the square around H. For $\hat{p} = K\hat{x}$ and $\hat{s} = (1 - \mathbf{1}^\top K)\hat{x}$, the hourly cost is

$$J(K) = \tfrac{1}{2\pi}\int \left[q|1 - \mathbf{1}^\top K|^2 + \textstyle\sum_i a_i |K_i|^2\right] \Phi_x \mathrm{d}\omega. \tag{A6}$$

Its integrand is $q - 2q\Re(\mathbf{1}^\top K) + K^* M K$. As $MH = q\mathbf{1}$ by (5) and $H^* M H = q(1 - H_{\mathrm{G}})$, it equals $qH_{\mathrm{G}} + (K-H)^* M (K-H)$, so the cost of K exceeds $J(H)$ by the M-weighted distance of K from H. For $p = Hx + \zeta$ with ζ jointly WSS with x, the terms linear in ζ vanish as $MH = q\mathbf{1}$, so

$$J = J(H) + \tfrac{1}{2\pi}\int \mathrm{tr}(M\Phi_\zeta)\,\mathrm{d}\omega, \tag{A7}$$

and $\zeta = (K - H)x$ gives (6). *Gap (7).* With $D = \mathrm{diag}\{2c_i^\tau\}$, the dispatch is $p = D^{-1}(\lambda\mathbf{1} - \tilde{\mu})$. Substituting it into the balance $\mathbf{1}^\top p + \lambda/(2q) = x$ gives $\lambda = (x + \mathbf{1}^\top D^{-1}\tilde{\mu})/(\mathbf{1}^\top D^{-1}\mathbf{1} + 1/(2q))$, so p is affine in $\tilde{\mu}$ with slope Π. At $\tilde{\mu} = \mu$, it is the hindsight optimum Hx, so with the prediction error $\delta = \tilde{\mu} - \mu$,

$$p = Hx + \Pi\delta. \tag{A8}$$

Substituting $\zeta = \Pi\delta$, with $\Phi_\zeta = \Pi\Phi_\delta\Pi^\top$, into (A7) and multiplying by T gives (7). *Floor $T\Delta^\star$.* First, M is rational in ω^2, bounded, and $M \succeq (\min_i c_i^\tau)I$, so $M = V^*V$ with $V^{\pm 1}$ stable. The rational integrable Φ_x equals $|\varphi|^2$ with φ outer in the Hardy space $\mathcal{H}^2$ of causal finite-energy signals. Let $W = V\varphi$, so by (6), $J(K) - J(H) = \frac{1}{2\pi}\|WK - WH\|^2$. Then, for a causal stable K, WK is causal and thus orthogonal to the anticausal part $\{WH\}_-$. Splitting $WH = \{WH\}_+ + \{WH\}_-$ gives

$$J(K) = J(H) + \Delta^\star + \tfrac{1}{2\pi}\|WK - \{WH\}_+\|^2, \tag{A9}$$

so no causal LTI policy has a gap below $\Delta^\star$ per hour, or $T\Delta^\star$ over T hours. Finally, the bound is tight: WK ranges over $V(\mathcal{H}_\infty\varphi)^N$, which is dense in $(\mathcal{H}^2)^N$ (Beurling) as $V^{\pm 1}$ is stable, and if φ has relative degree one and no imaginary-axis zeros, $K^\star = W^{-1}\{WH\}_+$ is stable and proper and attains it. ∎

D. Proof of Corollary 2. We compute the variance of μ_i and the integral of its autocovariance R. For white LMP fluctuations of density S, μ_i has spectrum $S/(1+\tau_i^2\omega^2)^2$ by (A4), so

$$R(0) = \tfrac{S}{2\pi}\int \frac{\mathrm{d}\omega}{(1+\tau_i^2\omega^2)^2} = \frac{S}{4\tau_i}, \quad \int R(u)\,\mathrm{d}u = S. \tag{A10}$$

The integral correlation time is thus $S/R(0) = 4\tau_i$, and the mean of μ_i over $T \gg \tau_i$ hours has variance $S/T = R(0)/(T/4\tau_i)$, that of $T/(4\tau_i)$ independent samples. For a general LMP spectrum Φ_λ, the filter has unit gain at $\omega = 0$, so $\int R(u)\mathrm{d}u = \Phi_\lambda(0)$ and $T_i = \Phi_\lambda(0)/R(0) = \Phi_\lambda(0)/\mathrm{var}\,\mu_i$, as in Corollary 2. ∎

II. Benchmark Details

All benchmarks use the same network, fleet, and scoring as Algorithm 1; MPC, DQN, and cutoffs use 60 days of history. *1) MPC.* Every 30 min, MPC solves (1) over the next 24 h with the terminal penalty $\kappa_i(e_{i,t+24\,\mathrm{h}} - e_{i,0})^2$ and the forecast net load

$$\tilde{x}_{n,t+k} = d^1_{n,k} x_{n,t} + d^2_{n,k} x_{n,t+k-96} + d^3_{n,k} + r_{n,k}. \tag{A11}$$

Here, $d_{n,k}$ are least-squares coefficients for horizon k, and $r_{n,k}$ is an XGBoost correction from recent lags, the values 1, 2, and 7 days earlier, and the time of day, kept only if it lowers the error on the last 15 training days. The test RMSE is 24% and 53% of the net-load standard deviation at 15 min and 24 h ahead (62% for day-ahead persistence). *2) ADP.* Step 5 values $e_{i,t}$ at $v_i(e_{i,t-1}) = \max(v_i^0 + v_i^1 e_{i,t-1}, 0)$, $v_i^1 \le 0$, fitted by least squares to the hindsight $\mu_{i,t}$ at each monthly refit. ADP with features adds $\vartheta_i^\top f_t$ to v_i. *3) DQN.* Adapted from [5], each unit has a double deep Q-network (6-128-128-7) with the net load, its 1-h and 24-h averages, the SoC, and the time of day as state, seven power levels as actions, and minus the system cost as shared reward, retrained monthly on 7-day episodes for 10^5 steps, with the discount 0.995 tuned on the last 15 training days. A single-period ED tracks its setpoints. *4) Cutoffs.* From short to long duration, the units take the bands of x between trailing moving averages with windows tuned on the last 25 training days, plus an SoC correction. *5) Oracle.* Step 5 uses an affine fit of the exact averages (4) to the past $\mu_{i,t}$.